2.5 Modulus of a Complex Number

Just as the absolute value of a real number measures the distance of that number from origin along the real number line, the modulus of a complex number measures the distance of that number from the origin in the complex plane. Observe that the length of the line from the origin along the radial line to $z = x + iy$ is simply the hypotenuse of a right triangle, with one side of length x and the other side of length y.

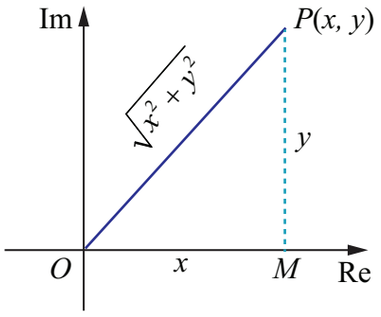


Fig. 2.16

Definition 2.4

If $z = x + iy$, then the modulus of z, denoted by $|z|$, is defined by $|z| = \sqrt{x^2 + y^2}$

For instance (i) $|i| = \sqrt{0^2 + 1^2} = 1$

(ii) $|-12i| = \sqrt{0^2 + (-12)^2} = 12$

(iii) $|12 - 5i| = \sqrt{12^2 + (-5)^2} = \sqrt{169} = 13$

Note

If $z = x + iy$, then $\bar{z} = x - iy$, then $z\bar{z} = (x + iy)(x - iy) = (x)^2 - (iy)^2 = x^2 + y^2 = |z|^2$.

$$|z|^2 = z\bar{z}.$$

2.5.1 Properties of Modulus of a complex number

(1) $|z| = |\bar{z}|$

(2) $|z_1 + z_2| \le |z_1| + |z_2|$ (Triangle inequality)

(3) $|z_1 z_2| = |z_1||z_2|$

(4) $|z_1 - z_2| \ge ||z_1| - |z_2||$

(5) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$, $z_2 \neq 0$

(6) $|z^n| = |z|^n$, where n is an integer

(7) $\mathrm{Re}(z) \le |z|$

(8) $\mathrm{Im}(z) \le |z|$

Let us prove some of the properties.

Property Triangle inequality

For any two complex numbers z_1 and z_2, we have $|z_1 + z_2| \le |z_1| + |z_2|$.

Proof

Using $|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2})$ $(\because |z|^2 = z\bar{z})$

$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$ $(\because \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2)$

$= z_1\bar{z}_1 + (z_1\bar{z}_2 + \bar{z}_1 z_2) + z_2\bar{z}_2$

$= z_1\bar{z}_1 + \left(z_1\bar{z}_2 + \overline{z_1\bar{z}_2}\right) + z_2\bar{z}_2$ $\left(\because \bar{\bar{z}} = z\right)$

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